



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani Pilani Campus

First Semester 2026-27 Instruction Division Course Handout (Part II)

Date: 01/08/2026

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No : **PHA G542**
Course Title : **Advanced Physical Pharmaceutics**
Instructor-in-charge : **Dr. Gautam Singhvi**
Instructor : **Vaibhavi Desai, Yogesh Khairnar, Rajat Vashist, Dhruv Sharma, Sakshi Lad, Adarsh Sharma**

1. Course Description:

Physicochemical principles involved in effective formulation development, including various aspects like, pharmaceutical preformulation including bulk characteristics, polymorphism, solubility analysis, solid and solution state stability, compatibility. This course also cover various ICH guidelines like stability study, specifications, impurities which are required during formulation development. This course also discussed details about mechanical properties of powders, principles of dissolution, in-vitro in-vivo correlation (IVIVC), polymer science, etc.

2. Scope and objective of the course:

Prior to the development of any dosage form with a new or old drug candidate, it is essential that certain fundamental physical and chemical properties of the drug molecule and other derived properties of the drug are determined. This information dictates possible approaches in formulation development. The aim of this course is to make the students understand those physicochemical principles which must be applied in the formulation and development of an efficacious dosage form, so that required duration and intensity of effect can be ensured at the intended site of action.

3. Text Book (TB):

TB 1: Carstensen, J. T, Pharmaceutical Preformulation. Technomic Publishing comp. USA.

4. Reference Books (R): R1: Carstensen, J. T., Rhodes, C.T., 2000. Drug Stability: Principles and Practices, Drugs and Pharm. Sci. Series, Vol. 43, 3rd edn, Marcel Dekker Inc., New York.



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R2: Martin's Physical Pharmacy and Pharmaceutical Sciences.

R3: ICH Q guidelines -http://www.ich.org/fileadmin/Public_Web_Site/ICH_Products/Guidelines/Quality

R 4: Sanford Bolton, Charles Bon, Pharmaceutical Statistics, Informa Healthcare, 5th Edition, 2010.

5. Course Plan

Module Number	Lecture session/Tutorial Session.	Reference	Learning Outcome
1. Introduction and significance of advanced physical pharmaceutics.	1. Introduction to physical pharmaceutics	Class notes	The student should be able to understand the importance and overview of this course.
	2. Overview of pharmaceutical product development.	Class notes	
2. Preformulation Concepts	1. Overview of preformulation studies carried out during/ prior to formulation development	Class notes. Chapter 1, TB	Students will be able to develop analytical method and understanding for various preformulation investigation and their role in product development.
	2. Rational approach to development and validation of stability-indicating analytical method	Q2, R3	
	3. Study and impact of polymorphism, hydrates, hygroscopicity; vapour pressure	Chapter 7, TB	
	4. Determination of solubility. Factors/ parameters affecting solubility	Chapter 2, TB; Chapter 9, R2	
	5. Impact of powder densities, particle size, particle size distribution on process and product performance	Chapter 11, TB; Chapter 18, R2	
	6. Compatibility test for solid and liquid dosage forms.	Chapter 14, R2	
3. Physicochemical aspects of pharmaceutical preparations	1. Important physicochemical/ mechanical aspects to be considered during design and formulation of good quality pharmaceutical preparations	Class notes,	Understanding of important physicochemical/ mechanical factors during design and formulation development.





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4. Drug and formulation stability	1. Stability studies of drug substances and drug products with special reference to ICH guidelines,	Q1, R3	Understanding of various regulatory guideline which mandatory to follow during product development.
	2. Specifications and impurity limits	Q3 and Q6, R3	
	3. Physical stability testing of pharmaceutical preparations		
5. Optimization techniques in Pharmaceutical Formulation	1. Concept and parameters of optimization, optimization techniques in pharmaceutical formulation and processing.	Chapter 9 and 16, R.4	Understanding about quality by design concept for optimization of formulation and process.
6. Surface phenomenon and Rheology concepts	1. Principles and applications of adsorption; Factors affecting surface tension and Pharmaceutical applications of surfactants	Chapter 9, TB; Chapter 15-16, R2	Understanding about surface adsorption at liquid interface and rheological behavior of liquid and semi solid formulations.
	2. Newtonian and Non- Newtonian flow characteristics and application	Chapter 20, R2	
7. Polymers types and functions	1. Various types of polymers employed in dosage forms and approach for choosing the right ones; Mechanism of release with respect to polymers	Class notes and research articles	Polymer nature and application
8. Diffusion and in-vitro dissolution methodology	1. Principles of diffusion and dissolution, Release characterization	Chapter 10, TB; Chapter 13, R2.	In-vitro dissolution methodology for pharmaceutical product development and strategy for simulation of in-vivo drug release.
	2. In-vitro drug release (dissolution) test design. Similarity factors – f2 and f1		
	3. In vitro - in vivo correlations (IVIVC) with respect to BCS classification		
	4. Linearity Concept of significance, Standard deviation, students T-test , ANOVA test.	Chapter 7 and 8, R.4	
9. Recent advancement	1. Recent journal articles for discussion	Journal articles	Recent update in pharmaceuticals.





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6. Evaluation Scheme:

Components	Duration	Weightage	Date and Time	Remarks
Mid-semester test	90 minutes	25%	as per time table	Closed Book
Seminar/Assignments/ Quiz	Continuous	20%	Continuous	Open book
Lab work	Continuous	20%	Continuous	Open book
Comprehensive examination	180 minutes	35%	as per time table	Closed Book

After completing this course the students will be able to understand

- The elements of preformulation studies.
- Various stages of characterization of active pharmaceutical ingredients and drug product development
- Physicochemical and mechanical factors of drug and excipients
- In-vitro dissolution methodology for product development

7. **Mid-Semester Evaluation:** Will be announced after Mid sem exam.

8. **Make-up:** Prior approval or intimation to take a make-up is a must. It is solely the discretion of the instructor-in-charge dependent upon the genuineness of the circumstances to allow a student to appear for a make-up evaluation component.

9. **Grading policy:** As specified in Handout – Part I, appended to the timetable, the instructor in-charge reserves the right to award a NC report in case the student does not make himself/ herself available for any of the evaluation component mentioned above. Also it is not imperative on part of the instructor in-charge to award all the grades. Borderline cases during grading will be judged on the basis of regularity to classes and consistency or progress in the performance in evaluation components.

10. **Important Note:** It is required to get 40% of the median marks of the class to clear this course. **If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC.**

11. **Chamber Consultation Hours:** To be announced in the class.

12. **Notices:** Notices concerning the course will be informed in the regular class.

Instructor-in-charge

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